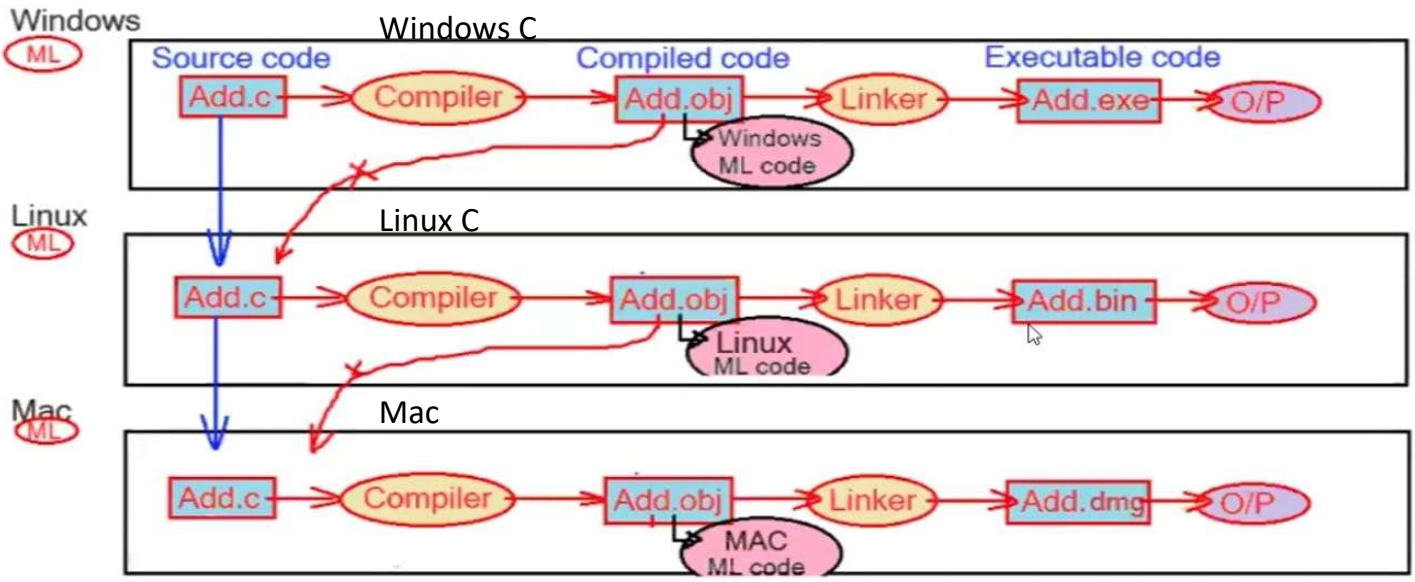
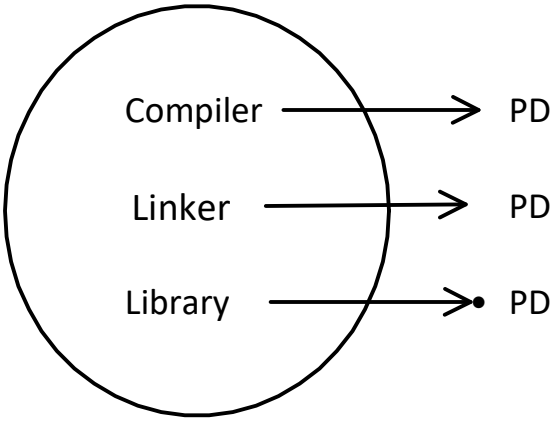
C, C++ programs are PD **(ML --> Machine Language)**

- In C and C++ languages, programs are compiled targeting to one OS.
- Hence the compiled code <u>contains ML</u> of a particular OS directly.
- One OS machine language format is not understandable to another OS
- Hence the C, C++ programs compiled to cannot be executed by other OS.
- Hence the C, C++ programs are platform dependent.

Below diagram shows C and C++ programs compilation, execution and PD problem

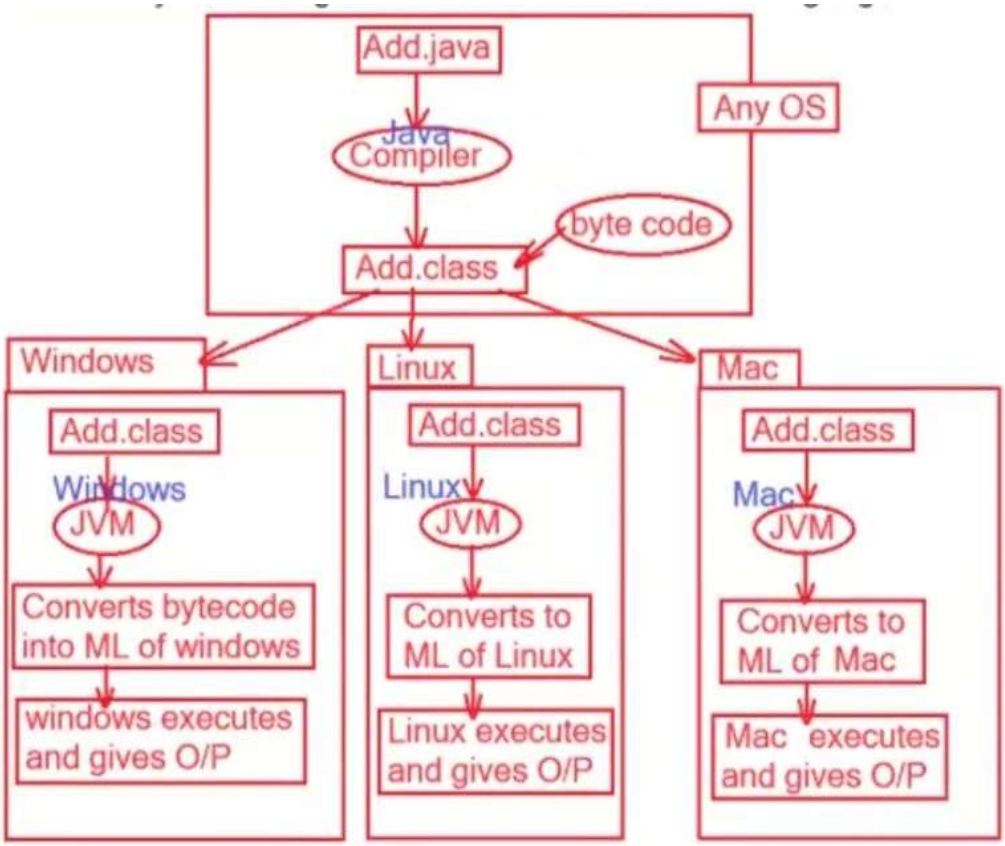


C S/W → PD



- Not only C program but also C s/w is platform dependent, because all its components compiler, linker and library are platform dependent
- So, we will have separate C s/w for every OS.

How Java achieved Platform Independency PI ?

- In Java, programs are not compiled into ML of particular OS.
- It means in Java, programs are not compiled targeting to one OS.
- In Java, programs are compiled into one common languages code called byte code then later that byte code is converted into ML of particular OS by JVM in which it is running. We have separate JVM for OS for converting the same bytecode in to a particular OS ML.
- Hence we can say, for achieving platform independency
  "In Java, Ml generation is moved from compilation phase to execution phase by introducing new compiler software, bytecode and JVM."
  - The new compiler software is responsible for converting [source code to byte code]
  - Then byte code is an instruction set to the JVM for running a java program
  - The JVM is a Java platform it is responsible to run Java byte code in a PI ways by converting them in current OS machine language.

Below diagram shows Java platform independency